Resource-constrained Fairness

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Access to resources strongly constrains the decisions we make. While we might wish to offer every student a scholarship, or schedule every patient for follow-up meetings with a specialist, limited resources make this infeasible. When deploying machine learning systems, these resource constraints are typically enforced by adjusting the classifier threshold. However, these finite resource limitations are disregarded by most existing tools for fair machine learning, which do not allow for the specification of resource limitations and do not remain fair when varying thresholds. This makes them ill-suited for real-world deployment. Our research introduces the concept of "*resource-constrained fairness*" and quantifies the cost of fairness within these constraints. We demonstrate that the level of available resources significantly influences this cost, a factor overlooked in prior evaluations.

Keywords: Algorithmic Fairness, Bias Mitigation, Machine Learning, Responsible AI in Practice

Reference Format:
Sofie Goethals, Eoin Delaney, Brent Mittelstadt, and Chris Russell. 2026. Resource-constrained Fairness. In *Proceedings of Fifth European Conference on Algorithmic Fairness (ECAF'26).* Proceedings of Machine Learning Research, 25 pages.

1 Introduction

Machine learning models are used to make decisions in many high-impact areas of our lives such as finance, justice, and healthcare [44]. Fair machine learning has emerged in response to the notion that simply making maximally accurate decisions is not enough and that training high-performance classifiers can result in both the transfer of existing biases from data to new decisions, as well as the introduction of new biases [58]. Many studies that focus on improving fairness in machine learning overlook the practical limitations under which these models operate. For example, scenarios including university admissions, healthcare provision, and corporate hiring, are usually constrained by finite *resources*. Universities have a restricted quota of students to admit annually, healthcare facilities are bounded by available space and staff, and companies have a limited number of positions to fill. Even for banking, where in principle banks can keep making loans providing enough people pay them back, for any particular time period they will have a limited set of resources and only be able to loan out a certain amount. In all these cases, *once the available resources are fully used,* increasing the selection rate of disadvantaged groups must necessarily involve reducing selection from more advantaged groups.

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ECAF'26, September 02–September 04, 2026, Ghent, BE

However, limited resources are not taken into account in most discussions in fair machine learning. For example, later we demonstrate that the labels produced by some well-known bias mitigation methods do not correspond to a fixed selection rate (Figure 5, Appendix C). Kwegyir-Aggrey et al. [37] highlight how practitioners typically need to adapt the threshold to ensure that outcomes meet their domain-specific needs, while Corbett-Davies et al. [15] argue that the trade-offs in fair decision-making are most acute when there are constrained resources. The inability to account for real-world constraints may contribute to the tiny number of fair algorithms publicly being deployed. This contrasts starkly with the abundance of studies that use fairness metrics solely for the evaluation of deployed systems [6, 10].

In the typical setting of an unconstrained budget, Mittelstadt et al. [46] posit that increasing harm to advantaged groups solely to achieve fairness is not an optimal approach and refer to this as '*leveling down*'. We agree with this viewpoint, but argue that within contexts constrained by limited resources, rebalancing is often necessary to redistribute the resources. In these circumstances, deliberately not fully utilizing resources would be a form of '*leveling down*'. In this study, we make the following practical and theoretical contributions:

- We formulate fair machine learning as a resource-constrained problem, where we treat positive decisions as a resource to allocate among different groups.
- We provide a connection to *leveling up* [46], showing that under constrained resources, leveling up results in a solution that enforces equality in harm between groups (Section 3) .
- We extend prior work on the cost of fairness [16] by quantifying this cost within resource-constrained settings. Using underlying parameters, we derive theoretical mathematical bounds for this cost (Section 4).
- We empirically validate these patterns on datasets from the fairness domain, and discuss some of the practical implications. In particular, we highlight the large influence of the resource level on the cost of fairness (Section 5).
- We explore the practical implications of our findings and discuss what strategic actions practitioners might undertake to mitigate trade-offs between groups when fairness is constrained by resource limitations.

This paper is structured as follows. We begin by reviewing related work and examining the theoretical cost of fairness, including mathematical bounds. We then turn to an empirical investigation, using a set of datasets commonly used in fairness research. The goal of this paper is not to propose a new bias mitigation method or to provide a benchmark of methods, but rather to shed new light on the actual cost of fairness in different contexts and the factors that influence it.

2 Background

We consider fairness in classification. Starting with a classifier $c(\cdot)$, we let $\mathrm{H}_{\mathscr{D}}[c]$ be a measure of expected harm of a classifier $c(\cdot)$ over a particular input distribution $\mathscr{D}$ where y_x represents the true target label of instance x, i.e.,

$$\mathrm{H}_{\mathscr{D}}[c] = \mathrm{E}_{x \in \mathscr{D}} H(y_x, c(x)) \tag{1}$$

Such measures of a harm $\mathrm{H}_{\mathscr{D}}[c]$ might be $1-$specificity (for example, when measuring the proportion of people incorrectly stopped by the police); or $1-$recall (when measuring the number of cancer cases unflagged for follow-up treatment).

In fair classification, we typically measure fairness with respect to a protected attribute, such as gender or ethnicity. Using this protected attribute, we can partition the dataset into groups. Group fairness metrics measure the (in)equality between these groups with respect to some measure of harm. For example, equal opportunity requires the recall between groups to be equal, while demographic parity enforces an equal selection rate [55]. Typically, a fair classifier is found by minimizing some global loss ℓ (such as accuracy or a continuous proxy such as the logistic loss) while ensuring that the harm is the same per group.

This means that we are searching for a solution to the following problem (where $\mathscr{G}$ is a partitioning of the distribution into groups with respect to a particular protected attribute, and w the weights of the classifier):

$$\begin{gathered}\min_w \mathrm{E}_{x\in\mathscr{D}}\ell(y_x, c(x)) \\ \text{such that } \mathrm{H}_{g_i}[c] = \mathrm{H}_{g_j}[c]\,\forall g_i, g_j \in \mathscr{G}\end{gathered} \tag{2}$$

Notions of fairness include demographic parity, where the harm corresponds to $1-$ the selection rate, and equal opportunity where it corresponds to the false negative rate.

2.1 Leveling down

Mittelstadt et al. [46] observe that methods to enforce fairness often level down; that is, they may enforce fairness by decreasing harm in some groups, but also by increasing harm to other groups (e.g. naively enforcing Equal Opportunity while detecting cancer will often result in a higher false negative rate for some groups – they receive a lower rate of cancer detection than they would otherwise). This process of enforcing fairness can alter the overall selection rate of the model [26], and is distinct from any leveling down that might be inherently required by the resource constraints discussed earlier. To this end, they suggested replacing Equation 2 with rate constraints, which enforced that the harm in question (e.g. being denied follow-up care) should be below a certain level h for every group:

$$\begin{gathered}\min_w \mathrm{E}_{x\in\mathscr{D}}\ell(y_x, c(x)) \\ \text{such that } \mathrm{H}_g[c] \leq h\,\forall g \in \mathscr{G}\end{gathered} \tag{3}$$

This is the same as saying that in the worst case, *the harm should be below h, for any group*.

The challenge now becomes "How should h be set?" When deploying a particular model, stakeholders and data scientists often have to agree on acceptable global levels of harm, for example, an acceptable recall rate (referred to as sensitivity in the medical literature) for early cancer detection. A similar process can also select a maximal per group harm.

This notion of leveling up and decreasing the harm in Equation 3 is related to but distinct from minimax-fair machine learning, which minimizes the error for the worst-off group [1, 18, 43]. In both minimax fairness and leveling up the concern is with reducing the harm to the worst-off group and only increasing harms to other groups when necessary, but in minimax fairness the harm is exclusively assumed to be caused by a lack of accuracy or high log-loss (this leads to a formulation of minimize the maximal loss, hence minimax). In comparison, leveling up is concerned with decreasing other harms such as per-group recall or selection rate, which are distinct from accuracy. [1]

[1] Both notions are closely aligned to the philosophical principle of the maximin rule [51]. According to Rawls, resources should be distributed in a way that maximizes the benefits for the least-advantaged members of society. However, as a fundamental limitation, neither minimax nor leveling up considers what should be done under constrained resources.

Taking a step back, we can ask the questions: "*If H is a harm, why not set h to zero? If particular groups are being harmed by not being selected, why not select everyone and not bother with machine learning? Similarly, if people are being harmed by a failure to detect cancer, why not schedule everyone for follow-up testing?*"

There are multiple possible answers here. One concerns personal utility: harms are generally not one-sided. Failure to repay a loan can lead to bankruptcy for both the lender and customer, resulting in devastating personal consequences. Scheduling unnecessary medical tests is at best alarming, and in the worst case can result in death, depending on how intrusive the follow-up tests are. These considerations have been addressed in prior work on the cost of fairness: for example, Menon and Williamson [45] measure personal utility using accuracy; Bakalar et al. [3] study the impact of implementing fairness principles in real-world situations; and Jorgensen et al. [36] show that fairness mitigation can sometimes cause more harm than good, depending on the context.

Constrained resources provide another answer. Ideally, we would prefer to offer scholarships to every student and fast-track treatment for every patient. However, the real world often lacks the resources needed to achieve this. Understanding how fairness choices are limited and guided by resource constraints is thus crucial to move fair machine learning from the theoretical to a production environment.

Fair classification under limited resources is related to the domain of fair allocation within welfare economics, which focuses on ensuring that resources such as time or physical goods are distributed among actors in a way that meets certain criteria [4, 8, 9, 19, 34, 50, 53]. Elzayn et al. [22] describe a resource allocation problem as a problem where there is a limited supply of resources to be distributed across multiple groups with different needs, such as allocating loans or disaster response. Bachmat and Navon [2] make the connection between algorithmic fairness definitions and resource allocation definitions, and examine the trade-offs between fairness and utility. Another related area is fair ranking, which aims to ensure fairness with respect to a top-k selection [60], and social welfare optimization [5, 14].

3 Constrained Resources

We formalize the notion of *resources*, as the proportion of instances predicted as positive by a machine learning model. This term can be used interchangeably with *capacity*, or with *selection rate*, *positive decision rate*, or *positive prediction rate* when talking about the proportion of positively predicted instances. We first focus on monotonic decreasing harms that start at a common value of 1 indicating maximum harm to every group, if no one is selected. For example, if the harm is $1 -$ recall, we expect the harm to be monotonic decreasing as more members of the group are selected. We will later consider monotonic increasing harms that start at a common value of 1 if everyone is selected.

Proposition 1. *Under constrained resources, maximally leveling up yields a solution in which harm is equal across groups, provided harms are monotone and continuous with respect to the group selection rate.*[2] [3]

PROOF. We write r as the selection rate of a classifier over a distribution $\mathscr{D}$. We consider the problem of optimizing the distribution of limited resources, to minimize the worst harm experienced by any group. To express that the harm per group is a direct function of the group rate, we drop c and write $H_i(r_i)$ for the harm to group i,

[2]Under unconstrained resources, this holds trivially.

[3]This solution is unique if the harms are strictly monotone at the solution found.

when the selection rate is r_i. This optimization problem is formulated as:

$$f_k(r) = \min_{\substack{\forall r_i \in [0,1] \\ \sum_{i \leq k} r_i \leq r}} \max_{i \leq k} H_i(r_i) \tag{4}$$

To simplify, we allow the assignment of excess capacity to a group, for example, some group g might only occupy 10% of the population but receive 20% of the capacity. Here, the harm remains monotone, but constant after the first 10%.

Case 1: Harm is monotonic decreasing with respect to r_i. We first consider two groups and evaluate two cases. One where all capacity r is assigned to group 1, and a second where all capacity is assigned to group 2. In the first situation,

$$H_1(r) - H_2(0) \geq 0 \tag{5}$$

And in the second

$$H_1(0) - H_2(r) \leq 0 \tag{6}$$

We linearly interpolate between these two solutions and consider a function l defined as:

$$l(\lambda) = H_1(\lambda r) - H_2((1-\lambda)r) \tag{7}$$

As $l(0) \leq 0$, and $l(1) \geq 0$, and l is continuous there must be some λ such that $l(\lambda) = 0$, and $H_1 = H_2$. Writing $r_1 = \lambda r$ and $r_2 = (1-\lambda)r$, this solution satisfies $r_1 + r_2 = r$, and is a minimizer of $\max(H_1(r_1), H_2(r_2))$, subject to the constraints. This follows as H_1 and H_2 are both monotonic, the only way to decrease the max is to increase both rates, which is not possible without violating $r_1 + r_2 \leq r$.

For $k \geq 3$ groups, we note:

$$f_k(r) = \min_{\substack{\forall r_i \in [0,1] \\ r_k + r_* \leq r}} \max\left(f_{k-1}(r_*), H_k(r_k)\right) \tag{8}$$

and that as f_{k-1} is monotonic decreasing[4] and $f_{k-1}(0) = 1$, we can recursively apply the two group proof. This gives us a solution where $H_i(r_i) = H_j(r_j) \forall i, j$ which is necessarily a minimum by the same argument as the two group case. Finally, we note that this minimum is unique if the functions H are strictly decreasing in a neighborhood around the solution found.

Case 2: Harm is monotonic increasing with respect to the rate. Now consider the case where harm is monotonic increasing with respect to the rate, e.g. Harm = 1 – specificity. In this case, the optimal solution is trivial with an overall selection rate of 0, as this would minimize harm across all groups. Equality is (trivially) achieved as no group receives any resources. However, in this case, it might be more realistic to flip the sign and require the overall selection rate to be at least some r. We can simply replace the classifier c with its negation, and now find that the harm is monotonic decreasing with respect to the selection rate of the new classifier. Revisiting the previous proof, we find that equality holds.

Hence, in all four cases considered (monotonic increasing or monotonic decreasing harm, and a global selection rate that is either constrained above or below some value) enforcing equality at a selection rate finds a solution for minimizing harm to the worst-off group under the same global rate constraint. □

[4]As r^* increases the constraint relaxes and the minimum found must not increase.

3.1 Cost of fairness

Before we proceed further, we note that assigning a cost to actions is inherently a political action, and that often these costs reflect the beliefs of data scientists and other professionals as much as they do the raw data. This is particularly apparent in the case of personal loans. While banks have good knowledge of the repayment habits of people they have lent money to in the past, they have much less knowledge of how people they did not lend money to might repay. In no small part, this knowledge asymmetry has allowed for the persistence of institutional redlining whereby loans are routinely denied to people in Black or Hispanic dominant neighbourhoods when they would be granted to individuals in similar circumstances who live in White dominant neighbourhoods [42]. Similarly, while it is tempting to assume that medical data is reliable, and for example, that we are forecasting the result of a future biopsy that will infallibly determine if cancer is present or not, for many problems the information is less clear-cut, and ground-truth may come from two experts attempting to diagnose from limited post-mortem records, with a third expert adjudicating in the case of disagreement [39].

Indeed, one justification for constraints such as demographic parity is that ground-truth data is often gender- or ethnically-biased, and in some circumstances we can *a priori* expect uniform rates across these populations (in line with the "We're All Equal" world-view of [24], which asserts that there are no innate differences between groups). As such, any estimate of cost comes with the usual caveat of *assuming the ground-truth is correct.* But even as simplified approximations, these costs remain useful for understanding the potential trade-offs in enforcing fairness. With this in mind, we consider the following question: *What is the global change in harm from an optimal classifier when we minimise the harm of the worst-off group?*

Under constrained resources, this has the same result as enforcing equality of harms between groups. We can measure this by the difference between the harm of the optimal classifier $c_w^o(.)$ and the classifier that satisfies fairness $c_w^f(.)$.

$$\mathrm{H}_{\mathscr{D}}[c_w^f] - \mathrm{H}_{\mathscr{D}}[c_w^o] \tag{9}$$

While other studies have previously analyzed the cost of fairness [16, 25, 28, 56], they did not consider the implicit trade-off that comes from constrained resources in decision-making, but instead measured how classifiers deteriorate with fairness. Some of the calculated costs may very well arise from selecting a different *number* of people; however, we will keep this number constant and focus instead on the costs associated with selecting a different *set* of individuals.

We might also be interested in performance metrics outside the fairness metric enforced. Of particular interest is the change in precision that occurs if we minimize the harm to the worst-off group. This represents a common scenario. In a medical context, it corresponds to the question: *What proportion of healthy rather than sick patients will we see as we increase test sensitivity for disadvantaged groups?*

This leads to a follow-up question: *What is the increase or decrease in selection rate needed to preserve the current rate of global harm, if we enforce equality?* The last question represents a common political solution to this problem. When particular groups are disadvantaged by the status quo, it is often easier to increase the resources, and target them at the disadvantaged groups, rather than requiring currently advantaged groups to accept less access to resources. The number of available resources can be an alterable parameter that should explicitly be taken into account.

4 The Theoretical Cost of Fairness

4.1 Bounding the cost of fairness

We now formalize these costs and consider the cost of the change induced by enforcing fairness at a particular selection rate. As we are modeling the cost of inducing fairness under rate constraints, we consider the changes in the labeling induced by swaps, where the label assigned to one instance changes from positive to negative, and the label assigned to another instance changes from negative to positive [57]. A sequence of swaps must always preserve the global selection rate, and any pair of labellings with the same selection rate can be transformed from one to another by a sequence of swaps. We write p for the proportion of instances in the dataset swapped from positive to negative (or equivalently, the proportion swapped from negative to positive), and c for the average cost of swapping a pair.

By definition, the cost of inducing fairness is:

$$\text{cost} = p \cdot c \tag{10}$$

Note that $p \in [0, 0.5]$, and for standard linear measures (e.g. accuracy, recall, specificity) the second term is bounded by

$$c = \frac{\text{\#instances in dataset}}{\text{\#instances used in the measure}} \tag{11}$$

i.e. $c = 1$ for accuracy, and, writing b for the label base rate, $c = b^{-1}$ for Recall and False Negative Rate, and $c = (1-b)^{-1}$ for Specificity and False Positive Rate. As such, pc and $c/2$ are both upper bounds on the cost of fairness.

For example, at a given global selection rate r, the cost of fairness, in terms of change in accuracy, is at most the proportion of swaps (p); while the cost in terms of decrease in recall is at most p/b.

We can create successive bounds based on this identity: Given r, p is bounded above by r (all positive instances are swapped); and $(1-r)$ (all negative instances are swapped). Therefore, the cost of fairness is bounded above by rc, and $(1-r)c$. As such, if either r or $(1-r)$ is much smaller than c^{-1}, there is little cost in enforcing fairness.

This matches our empirical findings in Figure 2 that the cost is typically bell-shaped with respect to the global selection-rate. The highest costs occur when the selection rates are closest to the selection rates of an unconstrained classifier. This means that many of the existing works on the cost of fairness [25, 33, 45] overestimate the cost of fairness for scenarios with other resource levels.

For scenarios where the selection rate substantially exceeds the base rate, this is because for any informative classifier, the majority of candidates that might be positive would be selected early on, regardless of which group they belong to, and the remaining pool of candidates are likely to have negative labels regardless of which group they belong to.

Similarly, given g, the proportion of the dataset occupied by the smallest group, we know $p < g$, and the cost of inducing fairness is bounded above by gc.[5] Again, if g is much smaller than c^{-1}, there is little cost in inducing fairness. This finding is supported in our experiments (see Figures 1g and 1h).

These bounds are much weaker for non-linear measures such as precision, if they exist. For precision, assuming a fixed global selection rate r, we have $c = r^{-1}$. Consequently, the low selection-rate bound given by $rc = 1$ is

[5] Relabelling only a subset of the smallest group is not always sufficient to enforce fairness, e.g. consider equal precision, but it holds for linear measures.

vacuous. However, other bounds remain informative when there is a higher selection rate – particularly, the bounds $(1-r)c$ and gc indicate that we should see similar patterns in the costs with respect to precision for higher selection rates. Empirically, we find this — precision exhibits similar trends to the linear measures of recall and accuracy. Reflecting the much weaker bound for low selection rates, it sometimes exhibits strong changes at low resource levels, which are not present for accuracy and recall. This can be seen in Figures 2b and 2d.

Finally, looking at the second term c (the average cost of relabeling), we find that this second factor explains much of what we will see in Figure 1. For example, increasing *global* noise decreases performance for both groups, at broadly similar proportions. This overall decrease shrinks the gap between the per-group performance, reducing the average cost of relabelling points. Similarly, increasing classification noise for the group with greatest classification error, or increasing the difference in base rates, increases the performance gap between groups and consequently the cost of relabelling.

Following this analysis, it is reasonable to ask if bounds exist for *leveling up*, and if there are bounded increases in selection rate for a disadvantaged group guaranteed to remove unfairness (while keeping the selection rate of the advantaged group as-is). In brief, the answer is yes for demographic parity; for equal opportunity you may need to relabel all datapoints in a disadvantaged group; and leveling up need not be possible for equal precision. [6]

For demographic parity, there are no stability concerns. Let g be the proportion of the dataset corresponding to the disadvantaged group, and r the global classifier selection rate. Then an upper bound for the selection rate excluding members of the disadvantaged group is $r/(1-g)$, and at most we require an additional $gr/(1-g)$ proportion of points to be positively labelled to obtain parity. For equal opportunity, only instances assigned a positive ground-truth label alter the recall, and it is possible that a badly-trained classifier selects all negatively labelled instances first, and only then positively labelled points. If we consider a classifier like this, and a group with only one positively labelled point (in the ground truth), all points within the group must be selected to get any non-zero recall rate for that group. As such, g is a bound. For equal precision, we consider the same example, and note that while you must select all datapoints to obtain non-zero precision, the precision will be the base rate of the group, and it may be insufficient for equality with respect to the other group.

4.2 Varying global selection rates

Strategic adjustments in selection rate may be necessary depending on the outcomes of a cost-benefit analysis. Until now, we assumed the resources to be fixed, but practitioners may find it feasible to increase or decrease them, particularly if the resource levels were initially determined under an unfair model allocation. For instance, in healthcare, additional investments could be directed towards expanding screening facilities to counterbalance the decrease in true positives, effectively maintaining the detection rates prior to implementing fairness measures. Similarly, in educational settings such as student admission, institutions might adjust the size of admitted cohorts, either decreasing it to preserve the average quality, or increasing it to have the same number of graduates. Donahue and Kleinberg [19] discuss how increasing the level of available resources is a critical goal where advocacy and political action can play a key role.

[6]These results differ from the previous paragraphs because the previous analysis was advantaged by the stability of measures – we altered per group selection rates and hoped that global measures remained stable. In this new case, we are trying to alter per-group measures and the stability of measures is now a disadvantage.

Another approach could involve refining the precision for the protected groups by collecting more data or enhancing data quality, albeit at an initial cost. This investment might reduce the overall fairness cost, yielding a more profitable model over time. However, high quality data may simply not be available for all groups or infeasible to collect for a variety of legal, political, and practical reasons, meaning it cannot be assumed that collecting more or “better" data will necessarily solve inequalities in resource distribution [48, 49, 58].

5 The Empirical Cost of Fairness

Our empirical analysis is organized around three questions:

- *RQ1: How does the level of available resources (the selection rate) affect the cost of fairness?*
- *RQ2: How is this cost affected by dataset characteristics; specifically base-rate disparity, the size of the disadvantaged group, and label noise?*
- *RQ3: How does the cost compare across fairness metrics, namely demographic parity and equal opportunity?*

We address RQ1 in Section 5.4, RQ2 in Section 5.3, and RQ3 in Section 5.5, after first reporting general results across all datasets (Section 5.2).

5.1 Experimental setup

5.1.1 Materials

We use several real-world datasets common in tabular fair machine learning [38]. We also extend our findings to two datasets (CelebA [41] and Fitzpatrick17k [27]) from computer vision. The **Adult Income** dataset is taken from the 1994 census data, and contains a prediction task identifying if an individual’s annual income surpasses $50,000. The **COMPAS** dataset gathers demographic details and criminal records of defendants from Broward County to forecast the likelihood of reoffending within two years. The **Dutch Census** dataset from 2001 captures aggregated demographic data in the Netherlands, utilized to determine if an individual’s occupation falls into a high-level (prestigious) or low-level category. The **Law Admission** dataset contains data from a 1991 survey by the Law School Admission Council (LSAC) across 163 U.S. law schools, aimed at predicting a student’s success on the bar exam. The **CelebA** dataset comprises over 200,000 celebrity images, each annotated with 40 attribute labels ranging from hair color to emotions, along with 5 landmark locations. We use the attribute ‘wearing earrings’ as target label, as it is highly skewed towards the non-Male class [59] and thus has a significant base rate disparity. The **Fitzpatrick17k** dataset consists of approximately 17,000 dermatologist-curated skin lesion images, categorized across the Fitzpatrick skin type scale. We preprocess the data and binarize the labels following the approach of [62]. We specify the protected and target attributes in Table 1. We also report the **base rate disparity** for each dataset, which is defined as the actual difference in the proportion of positive outcomes between groups in a dataset. If we consider a binary sensitive attribute A, with ns representing the disadvantaged group and s the advantaged group it can be mathematically expressed as:

$$P(Y = 1 \mid A = ns) - P(Y = 1 \mid A = s)$$

5.1.2 Methods and Metrics

We use a standard train-test split, where the model is trained on the training set and the predictive performance and cost of fairness is evaluated on a separate test set.

Table 1. Used datasets. The values between brackets represent the percentage of the dataset with that value.

Name	# instances	# attributes	Protected attribute	Disadvantaged group	Target attribute	Base rate disparity
Adult (US census)	48,842	10	Gender	Female (33.15%)	High income (23.93%)	19.46%
COMPAS	5,278	7	Race	African-American (60.15%)	Low risk (52.12%)	24.60%
Dutch (Dutch census)	60,420	11	Gender	Female (50.10%)	High occupation (47.60%)	29.86%
Law admission	20,798	11	Race	Non-White (15.90%)	Pass the bar (88.98%)	19.82%
CelebA (image)	202,599	NA	Gender	Male (38.65%)	Wear earrings (20.66%)	29.84%
Fitzpatrick17K (image)	16,012	NA	Skin color	Black (31.79%)	Skin cancer (13.65%)	3.99%

For the tabular datasets, we use eXtreme Gradient Boosting (XGBoost) [13]. We optimize the number of boosting rounds through 5-fold cross-validation on the training set with early stopping after 10 rounds if no improvement is seen. The optimized model is trained on the entire training set and evaluates the probability of positive class outcomes. For the image datasets, we employ a Resnet-50 (CelebA) and Resnet-18 (Fitzpatrick-17k) backbone [31] pretrained on ImageNet [17] for feature extraction. More details about the used parameters are provided in Appendix A. We also provide more information about the used performance metrics (AUC score, precision at the top R, recall at the top R, and accuracy at the top R) and fairness metrics (demographic parity, equal opportunity) in Appendix A.

5.1.3 *Enforcing fairness*

To enforce fairness (both demographic parity and equal opportunity), we apply a simple groupwise thresholding method that adjusts the decision threshold separately for each subgroup defined by the protected attribute. This method is a *post-processing* bias mitigation strategy, which is a class of methods that attempt to make the output of a machine learning model more fair *after* the model has been trained [44]. Our method is in line with the approach of Hardt et al. [30] to achieve equalized odds, but we implement it for demographic parity and equal opportunity.

This approach allows us to enforce fairness perfectly for every selection rate. We conduct an analysis across a range of selection rates r (from 1% to 100 %) representing different resource levels. This process is particularly straightforward for demographic parity. In this case, the harm corresponds to $1 -$ selection rate, and the process simplifies to just selecting the top r from each group.[7] We provide a comparison with other bias mitigation methods in Appendix C.

In addition to post-processing bias mitigation methods, there are also methods that operate at different stages of the machine learning pipeline. Specifically, pre-processing methods modify the data representation before model training, while in-processing methods incorporate fairness constraints or regularization directly into the learning algorithm [44]. Although a comparison across bias mitigation methods is not the goal of this paper, we provide an illustration of how some other bias mitigation methods behave across resource levels. We use three classical bias mitigation methods from AIF360 [7], namely Disparate Impact Remover (DIR) [23], Adversarial Debiasing [61]

[7]For demographic parity, ideal thresholds can be calculated on the test set, but for most other measures the thresholds need to be calculated on a separate validation set, as they require access to the target label in order to compute the harms. Owing to sampling error, thresholds selected on the validation data do not correspond perfectly to selection rates on unseen data. As such, to ensure that the number of instances predicted as positive exactly matches the available resources, we calculate the proportion of resources to allocate to each group for each resource level R on the validation set, and transfer these proportions to the test set.

and the Meta Fair Classifier [11]. DIR is a preprocessing method that attempts to make the model more fair by modifying the training data to reduce the influence of sensitive attributes, but preserving rank-ordering within groups. Adversarial Debiasing is an inprocessing bias mitigation method that combines a classifier that predicts the class label with an adversary that predicts the sensitive attribute, where the goal is to maximize the classifier's performance while minimizing that of the adversary. The Meta Fair Classifier is also an inprocessing method that takes the fairness metric as part of the input, and returns a classifier optimized with respect to that fairness metric. We use the default parameters. The goal of these methods is to make the final labels more fair, but as we will see in the results, they will not enforce fairness perfectly at every selection rate. [8]

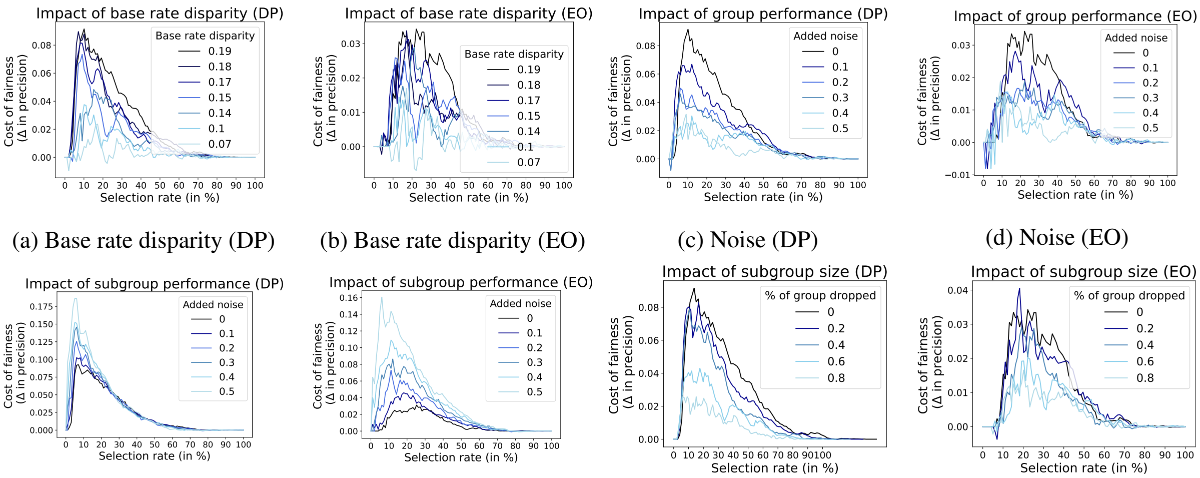
5.2 General results

Table 2. Model performance (entire population, advantaged group and disadvantaged group) and the average cost of fairness (loss in precision, recall and accuracy) when enforcing DP and EO. The average is calculated over all the selection rates from 1% to 100%.

Dataset	Adult	COMPAS	Dutch	Law	CelebA	Fitz17K
AUC	0.896	0.812	0.917	0.870	0.957	0.826
AUC_{priv}	0.887	0.797	0.884	0.847	0.931	0.829
AUC_{prot}	0.871	0.793	0.914	0.858	0.969	0.814
Avg. loss in precision (DP)	0.025	0.026	0.028	0.005	0.069	0.001
Avg. loss in precision (EO)	0.009	0.014	0.007	0.003	0.009	0.000
Avg. loss in recall (DP)	0.023	0.016	0.028	0.004	0.062	0.001
Avg. loss in recall (EO)	0.010	0.007	0.008	0.002	0.013	0.001
Avg. loss in accuracy (DP)	0.011	0.017	0.026	0.007	0.026	0.000
Avg. loss in accuracy (EO)	0.005	0.008	0.008	0.004	0.005	0.000

In Table 2, we analyze the AUC performance of a machine learning model for different groups: the entire population, and separately for disadvantaged and advantaged groups. Additionally, we report the average cost of enforcing fairness as the loss in precision, recall and accuracy when switching from the default allocation to the fair allocation, averaged over all selection rates. We see that the average cost of enforcing fairness is the lowest for the Fitzpatrick17K dataset, which also has a very low base rate disparity (see Table 1). The average cost is also very low for the Law dataset, despite its significant base rate disparity. A possible reason for this could be the relatively small size of the disadvantaged group in this dataset, which results in less reallocation of resources. The average cost of enforcing fairness across other datasets is more similar, however it is worth noting that the cost associated with enforcing both fairness metrics on CelebA is higher than for the other datasets. It is hard to attribute the difference in costs between the datasets to a single factor, as many of the parameters will be different. This is why we perform a separate analysis on the Adult Income dataset.

[8]The code for the experiments is publicly available here: https://github.com/SofieGoethals/RCF



(a) Base rate disparity (DP) (b) Base rate disparity (EO) (c) Noise (DP) (d) Noise (EO)

(e) Subgroup noise (DP) (f) Subgroup noise (EO) (g) Subgroup size (DP) (h) Subgroup size (EO)

Fig. 1. Impact of parameters on the cost of fairness, measured as the **loss in precision** (Adult Income dataset). We see that reducing the base rate disparity leads to a lower cost of fairness. Similarly, introducing more noise across all groups, also lowers the cost of fairness, but introducing more noise exclusively in the disadvantaged group, increases the cost of fairness. Finally, reducing the size of the disadvantaged group, results in a decrease in the cost of fairness as well.

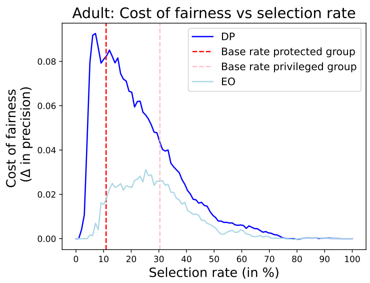
5.3 Impact of key parameters

Here, we systematically vary key parameters—one at a time—to observe their effects on the cost of fairness. Specifically, we explore modifications to the base rate disparity, the size of the disadvantaged group, the noisiness of the whole dataset and the noisiness of the disadvantaged group. The impacts of these changes are shown in Figure 1, with the original dataset's results represented by a black line.

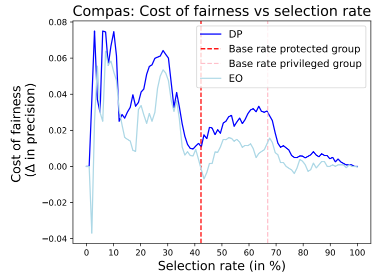
Base rate disparity First, we alter the base rate disparity. We see in Figures 1a and 1b, that a reduction in the base rate disparity leads to a lower cost of fairness. This trend is more pronounced when enforcing DP compared to EO, but it is evident under both fairness metrics. [45] also find that the trade-off between accuracy and fairness is determined by the strength of the correlation between the sensitive attribute and the target.

Noise We alter the performance of the model by introducing random noise (with varying degrees from 0 to 0.5) to the feature values of the whole dataset. The base rates of both groups remain the same, but the model struggles more with correctly classifying individuals, resulting in a lower AUC score. We see that as model performance decreases, the average cost of fairness also goes down, both for DP and EO (Figures 1c-1d). This global decrease shrinks the gap between the performances of each group, reducing the overall cost of fairness. This phenomenon might also explain why the average cost of fairness is so high for the CelebA dataset in Table 2. The model is very good at distinguishing negatives and positives from each other (as measured by the AUC), so the average cost will be high.[9]

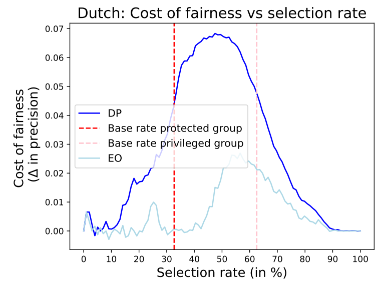
[9]Following this reasoning, we see that if we were to fit a perfect model (so the model predicts the target label perfectly for all instances) to a dataset with some level of base rate disparity, then the maximum cost of enforcing demographic parity would be reached at a point between the



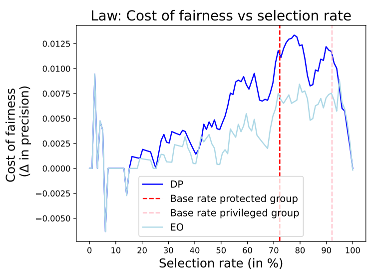
(a) Adult



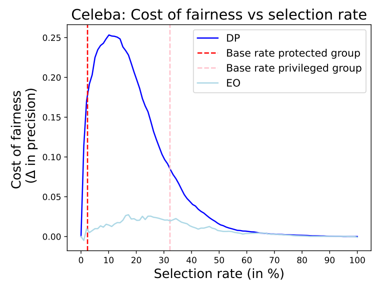
(b) COMPAS



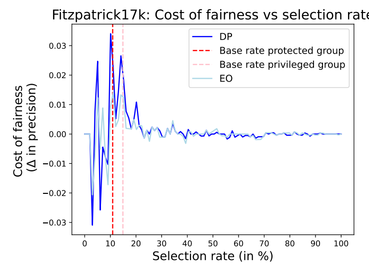
(c) Dutch



(d) Law



(e) CelebA - Earrings



(f) Fitzpatrick17k

Fig. 2. Cost of fairness (loss in **precision**) for all datasets when enforcing demographic parity (DP) and equal opportunity (EO).

Subgroup noise We can also add noise only to the members of the disadvantaged group. This lowers the performance of the model for the disadvantaged group, but not for members of the advantaged group. As shown in Figures 1e and 1f, this modification increases the cost of fairness for both DP and EO, with a larger effect for EO. [12] also find that when there is a difference in noise level, and available covariates are not equally predictive of the outcome in both groups, fairness cannot be satisfied without sacrificing accuracy. [20] confirm that if there is not enough separability information for one group compared to the other, being fair will reduce the accuracy.

Size of the disadvantaged group Lastly, the size of the disadvantaged group substantially influences the cost of fairness. When the disadvantaged group is smaller, fewer adjustments are necessary to achieve fairness, thereby reducing the cost. This relationship is clearly illustrated in Figures 1g and 1h, where reducing the size of the disadvantaged group, by randomly excluding part of this group, consistently lowers the cost of fairness. This is an expected finding and aligns with our bounds analysis in Section 4, however, we have not seen it discussed previously.

5.4 Impact of selection rate

Figures 2a-2f demonstrate the loss in precision for various selection rates when using a fair allocation compared to the default allocation of the machine learning model. We see that this cost varies substantially depending on

base rates and be exactly equal to the level of base rate disparity in that dataset. In the case of CelebA, the model is not perfect but the maximum cost is also relatively close to the level of the base rate disparity, and is reached at a point between the base rates.

the available resources. For the Adult (Figure 2a), COMPAS (Figure 2b), CelebA (Figure 2e) and Fitzpatrick 17K (Figure 2f) datasets, the cost is highest for low selection rates, while for the Dutch (Figure 2c) dataset, the cost is the highest for medium selection rates, and for the Law (Figure 2d) dataset, the cost is the highest for high selection rates. The influence of resource level on the cost of fairness has not been discussed before, yet it is a critical factor. Depending on the resource level, the cost can either be negligible or very high. The results for recall and accuracy are in line with precision and can be found in Figures 3 and 4. We can also partially agree with [52] who posited that for very constrained top-*k* settings, so when the selection rate is very low, the cost of fairness can be negligible. However, we only find this to be true when the base rate is a lot higher and the performance of the model is good enough. We get more insights into how these trade-offs behave for each resource level by analyzing the different scenarios in Appendix D.

5.5 Impact of fairness metric

Table 2 and Figures 2-4 show that the cost of fairness is generally lower when enforcing equal opportunity (EO) than demographic parity (DP) across all selection rates. This is because EO typically requires fewer instances to be “swapped." Liu et al. [40] also find that the selection rate enforced by EO is closer to the optimal, while [30] show that enforcing EO leads to a lower loss in profits than enforcing DP. However, EO can be costlier when ensuring an equal true positive rate imposes stricter constraints than an equal positive rate, for example when one group has lower data quality. In such cases, DP is less affected since it disregards prediction accuracy, whereas EO requires equally poor recall across groups, degrading overall performance. As shown in Figures 1e- 1f, the cost of enforcing EO rises significantly faster than the cost of enforcing DP when one subgroup becomes noisier.

6 Conclusion

This work addresses the practical concern of enforcing fairness with a limited budget or, equivalently, as the selection rate (or threshold) is fixed, a situation that often arises in domains where fairness is a critical concern such as healthcare, education or social services. We unify two previously distinct principles: leveling up [46] and equality of harm, and have shown that the solutions for each approach coincide under constrained resources. We have also shown how simple methods for setting per group thresholds can enforce fairness under these rate constraints, which makes actual deployment very easy.

Building on this, we empirically and theoretically quantify the cost of fairness in constrained settings. Empirically, we measure the cost of fairness under rate constraints – in terms of decreased performance, or as an increase in global harm; and we derive theoretical bounds for these costs that align with our empirical results. While the bounds are not intended to be tight, they are informative, and align with our empirical findings. This allows us to investigate the actual cost of fairness, rather than changes in performance metrics that are heavily influenced by a change in the overall selection rate [26]. Our findings are validated with comprehensive and rigorous experimental analysis using real-world datasets. Compared to other fairness works, our analysis more closely aligns with real-world social impact challenges, where resources are often fixed, and organizations need to have reasonable expectations about what costs to expect and why. Most prior works on this topic made abstraction of the actual decision-making context the problem is situated in, while we demonstrate that it has a very large influence on the cost; particularly the behavior of the classifier, the user fairness metric and the level of available resources. This work contributes to the discourse around AI governance and is especially relevant to those concerned with the distributional impacts of algorithmic decisions, such as policymakers, social scientists, and ethicists.

7 Limitations

Naturally, our work has several limitations which also pose opportunities for future research. We study the cost of enforcing fairness perfectly, but another interesting research direction could be to study how to achieve different optimal trade-offs between precision and fairness under different resource limitations. Additionally, as previously highlighted, our cost estimates are based on the assumption that the ground-truth is accurate. Yet, despite this simplification, they remain a useful tool for evaluating the potential trade-offs of enforcing fairness. Furthermore, we assume that the cost of resources is homogeneous. In areas such as social services or medical follow-up, these costs might be heterogeneous as well. Lastly, for our analysis, we assume access to the sensitive attribute(s), while in reality, legal and ethical considerations may impose constraints on obtaining or utilizing certain sensitive information [29, 32, 35, 47, 54]. Extending our work to intersectional fairness scenarios where multiple sensitive attributes are considered simultaneously would be a rich avenue for interdisciplinary research across law, ethics, economics, and computer science.

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